

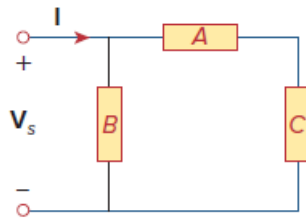
Problem

In the circuit of Fig. 11.71, device A receives 2 kW at 0.8 pf lagging, device B receives 3 kVA at 0.4 pf leading, while device C is inductive and consumes 1 kW and receives 500 VAR.

(a) Determine the power factor of the entire system.

(b) Find I given that $V_s = 120 \angle 45^\circ \text{ V rms}$.

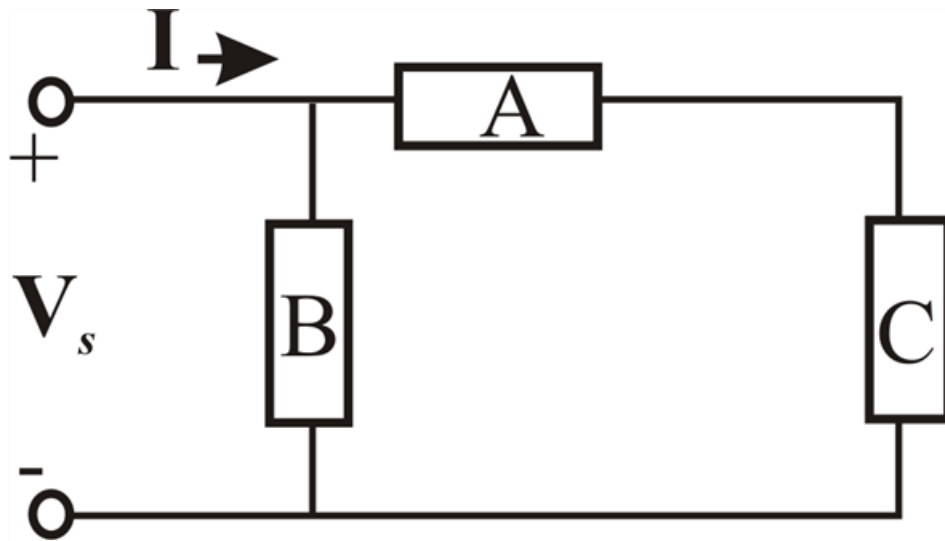
Figure 11.71:



Step-by-step solution

Step 1 of 8

Given figure is



Step 2 of 8

Given data

$$P_1 = 2000 \text{ W}, \cos \theta_1 = 0.8 \text{ lagging}$$

$$S_2 = 3000 \text{ VA}, \cos \theta_2 = 0.4 \text{ leading}$$

$$P_3 = 1000 \text{ W}, Q_3 = 500 \text{ VAR}$$

Step 3 of 8

We know that

$$S = \frac{P}{\cos \theta}$$

Then

$$S_1 = \frac{P_1}{\cos \theta_1}$$

$$S_1 = \frac{2000}{0.8}$$

$$S_1 = 2500 \text{ VA}$$

Step 4 of 8

then the complex power is,

$$S_1 = S_1 \angle \cos^{-1}(0.8)$$

$$S_1 = 2500 \angle 36.87^\circ \text{ VA} \dots\dots (1)$$

Step 5 of 8

Similarly

$$S_2 = S_2 \angle -\cos^{-1}(0.4) \quad (\because \text{Leading})$$

$$S_2 = 3000 \angle -66.42^\circ \text{ VA} \dots\dots (2)$$

And

$$S_3 = P_3 + jQ_3$$

$$S_3 = 1000 + j500$$

$$S_3 = 1118.03 \angle 26.56^\circ \text{ VA} \dots\dots (3)$$

Step 6 of 8

By adding (1),(2),(3) we get

$$S_T = S_1 + S_2 + S_3$$

$$S_T = 2500 \angle 36.87^\circ + 3000 \angle -66.42^\circ + 1118.03 \angle 26.56^\circ$$

$$S_T = 4266.50 \angle -10.119^\circ \text{ VA} \dots\dots (4)$$

Step 7 of 8

(a)

We know that power factor is,

$$PF = \cos \theta$$

$$PF = \cos(10.119^\circ) \quad (\text{from (4)})$$

$$\therefore \boxed{PF = 0.9844 \text{ Leading}}$$

Since the angle is Negative.

(b)

Given $\mathbf{V}_s = 120\angle 45^\circ \text{ V rms}$

We know that complex power,

$$\mathbf{S}_T = \mathbf{V}_s \mathbf{I}^*$$

$$4266.50\angle -10.119^\circ = (120\angle 45^\circ) \mathbf{I}^*$$

$$\mathbf{I}^* = \frac{4266.50\angle -10.119^\circ}{120\angle 45^\circ}$$

$$\mathbf{I}^* = 35.554\angle -55.119^\circ$$

$$\therefore \mathbf{I} = 35.554\angle 55.119^\circ \text{ A rms}$$